Marking Newton's comparison: classical widths and quantum probe costs

Two conditional recording theorems use different notions of error. In the classical independent-box model, a finite marking protocol distinguishing constant force from free motion requires $F^2\tau^3 > 9m\kappa$ when every resolution–recoil width product is at least κ. A specified balanced two-mark construction succeeds above $36m\kappa$. In the non-adaptive Gaussian model, tests invariant under unknown initial position and velocity instead obey the sharp bound $\tau\Delta E \geq 48z_{1-\epsilon}^2\kappa$. A quantum position-coupled probe supplies the **standard-deviation** product $\delta\Delta \geq \hbar/2$; its correlations change the Gaussian threshold. None of these statements derives physical $\hbar > 0$ from classical dynamics.

The hypotheses are integral to each theorem. Hard support widths, standard deviations, conditional posterior uncertainty and the hybrid-state disturbance spreads of general record bounds are separate quantities. This treatment incorporates the classical marking stages, body-independent-noise classification and Gaussian unread-record channel. Written model/coordinator checks are internal; the uncertainty and Gaussian channel ingredients are established, and no literature-priority claim is made.

Geometry and the independent-box record model

For mass $m > 0$, constant force $F > 0$ and duration $\tau > 0$, write

$$P(t) = \frac{Ft^2}{2m}, \quad s = P(\tau), \quad J = F\tau, \quad \tau\Delta E = Js = \frac{F^2\tau^3}{2m}, \quad A_{\text{inertial,fall}} = \frac{vF\tau^3}{6m}.$$

Thus $(3F/v)A = \tau\Delta E$, for horizontal speed $v > 0$. Unmarked polygon impulses, sagittas and areas vanish with mesh, with $\sum_j vF\tau_j^3/(6m) \leq vF\tau|\pi|^2/(6m)$ while $2mP(\tau_j)/\tau_j^2 = F$. No geometric inconsistency supplies a floor. The historical/action conventions are in Newton insertion.

Marks occur at $0 = t_0 < \cdots < t_k \leq \tau$. A datum is an interval of length $\delta_i > 0$ containing position; the initial momentum difference is bounded by Δ_0, and later impulse differences by Δ_j. The **model assumes the full independent adversarial feasibility box**: two histories share a record exactly when some $x \in \mathbb{R}$, $|w| \leq \Delta_0/m$ and $|r_j| \leq \Delta_j$ satisfy

$$|P(t_i) + d(t_i)| \leq \delta_i, \qquad d(t) = x + wt + \frac{1}{m}\sum_{j\geq 1} r_j(t - t_j)_+.$$

A statement about marginal widths alone does not imply this independent feasibility model if physical correlations remove allowed combinations. The last recoil is irrelevant to recorded positions. Assume $\delta_j\Delta_j \geq \kappa$ for every available mark.

Separation certificate. The protocol decides in the worst case iff some $\mu_0, \ldots, \mu_k$ with $\sum_i \mu_i = 0$ satisfy

$$\sum_i \mu_i P(t_i) > \sum_i |\mu_i|\delta_i + \frac{1}{m}\sum_j \Delta_j|S_j|, \qquad S_j = \sum_{i>j} \mu_i(t_i - t_j).$$

Indeed the feasible displacement set is the sum of a line and the image of a compact box, hence closed convex. Strict separation from the compact position-data box eliminates the free x by $\sum \mu_i = 0$. Minimizing independently over w, r_j gives exactly the displayed support function; conversely pairing a feasible history with the certificate contradicts it. Here $S_0 = \sum_i \mu_i t_i$ and $S_k = 0$.

Define the continuous piecewise-linear function

$$T(u) = \int_u^\tau \sum_{i:t_i>w} \mu_i \, dw.$$

It is constant before zero, zero after τ, has slope jump μ_j at t_j, and $T(t_j) = S_j$. Integration by parts gives

$$\sum_i \mu_i P(t_i) = \frac{F}{m} \int_0^\tau T, \qquad \sum_j \mu_j S_j = -E, \qquad E = \int_0^\tau T'^2.$$

The boundary terms vanish because $P'(0) = 0$, $T(\tau) = 0$ and T' is zero outside the interval. Termwise AM–GM bounds the certificate cost below by $2\sqrt{\kappa/m}\sum_j \sqrt{|\mu_j S_j|} \geq 2\sqrt{\kappa E/m}$. Also $|T(u)| \leq \sqrt{\tau - u}\sqrt{E}$, so $\int T \leq (2/3)\tau^{3/2}\sqrt{E}$. A strict certificate has positive $\int T$, and division proves

$$\boxed{F^2\tau^3 > 9m\kappa.}$$

If the model offers a preparation mark $\delta_0 = \sqrt{\kappa\tau/m}$, $\Delta_0 = \kappa/\delta_0$, and a terminal mark $\delta_1 \leq \sqrt{\kappa\tau/m}$, the inertial prediction interval has width $2\sqrt{\kappa\tau/m}$. It and its forced translate cannot share a terminal datum when $s > 3\sqrt{\kappa\tau/m}$, giving $F^2\tau^3 > 36m\kappa$. With this availability at every duration and $\kappa > 0$, the infimum deciding duration obeys

$$9m\kappa \leq F^2\tau_*^3 \leq 36m\kappa, \qquad \frac{9}{2}\kappa \leq \tau_*\Delta E(\tau_*) \leq 18\kappa,$$

$$\frac{3v\kappa}{2F} \leq A(\tau_*) \leq \frac{6v\kappa}{F}.$$

The last line corrects the factor-two area conversions in the earlier standalone derivation. A zero infimum of products **alone** does not imply arbitrarily short deciding protocols: the balanced widths may be unavailable, or a fixed minimum position resolution may remain. The zero-floor conclusion requires access to appropriately balanced arbitrarily small positive costs. The constants 9 and 36 are proved bounds, not an optimal threshold.

On a subwindow whose initial momentum is wholly unknown, the certificate additionally has $\sum_i \mu_i t_i = 0$. The same proof gives $F^2(t_b - t_a)^3 > 9m\kappa$ when only its own marks are used. This is a necessary condition for exhibiting force from that window, not a cessation of mathematical refinement or a sufficient decision criterion.

Determinate probes and the historical premise

For freely travelling classical probes with known incoming momentum, momentum conservation transfers uncertainty in outgoing momentum to recoil. Two recorded positions, separated by a freely chosen flight distance D, with bounded resolutions δ, δ_s, determine outgoing transverse momentum to width at most

$$p_\parallel(\delta + \delta_s)/D \longrightarrow 0.$$

This defeats an irrecoverable-recoil premise if those remote records are admissible. The infimum recoil width is zero; exact zero at a finite distance is not asserted. With a remaining minimum resolution Λ, the fixed-force time threshold need not vanish. In the simple two-mark construction, $s > 3\Lambda$ suffices after negligible recoil, giving an action scale $\sqrt{54mF\Lambda^3} \to 0$ as $F \downarrow 0$. The class and parameter being varied matter. The complete apparatus and calibration fibres supply separate finite mechanical tests of preparation and record assumptions.

Newton's *Opticks* supplies a measured interval of fits $\Lambda = 1/89000$ inch and a corpuscular hypothesis. The held companion contains Book II Part III Definition/Props. XII and XVIII, Book II Part I Obs. 13 and Queries 1 and 29 (passages). A historical reconstruction posits M1,

a corpuscular momentum scale p; M2, a gauge resolution at least Λ; and the **additional** M3, $\delta\Delta \geq \Lambda p$ even after subsequent records. Fits are longitudinal periods, not measured transverse standard deviations, and corpuscle size does not determine a momentum spread. M3 is therefore a modern record-level interpretation, not a theorem or explicit assertion of Newton. The remote-screen example shows why deterministic transport plus sufficiently rich readout does not derive it. A printed periodic disposition alone settles neither an operational uncertainty principle nor its negation.

For the original two-mark protocol, M3 gives prediction width at least $2\sqrt{\Lambda p\tau/m}$ and hence the necessary $\tau\Delta E > 8\Lambda p$. The three-mark mid-chord calculation instead gives $F^2\tau^3 > 64m\Lambda p$, correcting the old factor 128; neither is the general sharp protocol theorem. The modern substitutions $p = h/\lambda$ and $\Lambda = \lambda/2$ give $\Lambda p = h/2 = \pi\hbar$. They are supplied quantum identifications, not the probe standard-deviation minimum $\hbar/2$. Colour-dependent positive products need not have a positive infimum without a uniform bound; no cross-colour universality is inferred from bounded spectral range alone.

Quantum probe cost and body-independent noise

Supply canonical quantum kinematics, with physical $\hbar > 0$. For $U = \exp(-i\lambda\hat{y}\hat{P}_A/\hbar)$,

$$U^\dagger \hat{Q}_A U = \hat{Q}_A + \lambda\hat{y}, \qquad U^\dagger \hat{p} U = \hat{p} - \lambda\hat{P}_A.$$

Position and probe momentum are unchanged. Thus the **Heisenberg output** reading error and recoil are

$$\hat{N} = U^\dagger(\hat{Q}_A/\lambda)U - \hat{y} = \hat{Q}_A/\lambda, \qquad \hat{D} = -\lambda\hat{P}_A, \qquad [\hat{N}, \hat{D}] = -i\hbar.$$

The two commutators terminate the BCH series and prove these identities. For every joint state with finite relevant moments, Robertson–Schrödinger therefore gives

$$\delta^2\Delta^2 - c^2 \geq \hbar^2/4, \qquad \delta = \Delta\hat{N}, \quad \Delta = \Delta\hat{D}, \quad c = \mathrm{Cov}_{\mathrm{sym}}(\hat{N}, \hat{D}).$$

The coupling cancels. Pure Gaussian probes attain equality; c may be nonzero. These are operator spreads in a stated state, not an assertion that no subsequent record can estimate a realized kick. The general record bounds specify exactly which interpolating states are charged.

For any unitary mark and apparatus pointer $\hat{R}$, define $\hat{N} = U^\dagger\hat{R}U - \hat{y}$, $\hat{D} = U^\dagger\hat{p}U - \hat{p}$ and $\hat{X} = U^\dagger\hat{y}U - \hat{y}$. Expanding the output commutators gives

$$[\hat{N}, \hat{D}] = -i\hbar - [\hat{y}, \hat{D}] - [\hat{N}, \hat{p}].$$

If all three noises strongly commute with the body's Weyl operators, they are apparatus operators and $[\hat{N}, \hat{D}] = -i\hbar$, $[\hat{X}, \hat{N}] = [\hat{X}, \hat{D}] = 0$. When $\hat{X} = 0$, the coupling has the form

$$U = (1 \otimes U_0)e^{-i\hat{y}\otimes\hat{B}/\hbar}, \qquad \hat{D} = -\hat{B}, \quad \hat{R}_0 = U_0^\dagger\hat{R}U_0 = \hat{N}, \quad [\hat{R}_0, \hat{B}] = i\hbar.$$

Here commutation and canonical relations are in the regular strong Weyl sense, with the common domains needed for moments. To prove the form, U commutes with position and decomposes into measurable apparatus fibres $U(y)$. Exponentiating $U^\dagger\hat{p}U = \hat{p} + \hat{D}$ yields $U(y)^\dagger U(y+a) = e^{ia\hat{D}/\hbar}$, hence $U(y) = U_0 e^{iy\hat{D}/\hbar}$ up to choice of representative. The pointer identity then gives the displayed Weyl relation. Stone–von Neumann with multiplicity identifies a canonical probe factor $L^2(\mathbb{R}) \otimes \mathcal{K}$. Continuous unbounded canonical pointers are essential.

Even if $\hat{X} \neq 0$, translations generated by $\hat{D}$ shift $\hat{N}$ and fix both $\hat{D}$ and $\hat{X}$. This is why the recoil-only theorem in general bounds extends to this body-independent class, including

correlations across marks. Body-dependent noise needs the full displacement/disturbance theorem. Source parallels: Ozawa's independent-intervention setting (2003, metadata), Arthurs–Goodman (1988, metadata), and Davies's translation-covariant instrument framework (1976, metadata). No priority claim is made for the classification.

A compactly supported nonzero wavefunction cannot have compact Fourier support (Paley–Wiener). This obstructs simultaneous hard position/momentum support in the standard probe model. It does **not** by itself prove a universal assertion about the difference of noncommuting pre/post momenta for arbitrary instruments. The old passage from post-mark momentum tails to an infinite universally defined recoil width was unjustified. No quantum value is assigned here to the independent-box κ; quantum results use the explicit standard-deviation and operational formulations.

Sharp Gaussian invariant tests and their escape routes

Take non-adaptive marks with

$$R_i = y_0 + v_0 t_i + \theta P(t_i) + \xi_i + \frac{1}{m}\sum_{j<i} \iota_j (t_i - t_j), \qquad \theta \in \{0,1\}.$$

The pairs (ξ_j, ι_j) are centred Gaussian, independent across marks, with standard deviations δ_j, Δ_j and covariance $c_j = \rho_j \delta_j \Delta_j$. Require tests invariant under both unknown y_0, v_0. On their Gaussian quotient the optimal equal-prior error is $\Phi(-d/2)$, where

$$d^2 = \sup_{\sum u_i = \sum u_i t_i = 0} \frac{(u^T P)^2}{u^T \Sigma u}, \qquad u^T \Sigma u = \sum_j \left(\delta_j^2 u_j^2 + \frac{2c_j}{m} u_j S_j + \frac{\Delta_j^2}{m^2} S_j^2 \right).$$

Now S_j, T, E are the same certificate objects with $\mu = u$. The extra velocity invariance gives $T(0) = T(\tau) = 0$, and

$$(\int_0^\tau T)^2 \le \tau^3 E/12.$$

This follows by integrating against $g(t) = t(\tau - t)/2$, $-g'' = 1$: $\int T = \int g' T'$, $\int g'^2 = \tau^3/12$. If $\delta_j \Delta_j (1 - |\rho_j|) \ge \kappa_{\text{eff}}$, termwise AM–GM gives $u^T \Sigma u \ge 2\kappa_{\text{eff}} \sum |u_j S_j|/m \ge 2\kappa_{\text{eff}} E/m$. Consequently

$$d^2 \le \frac{F^2 \tau^3}{24 m \kappa_{\text{eff}}}, \qquad \tau \Delta E \ge 48 z_{1-\epsilon}^2 \kappa_{\text{eff}}.$$

For independent error and recoil, $\kappa_{\text{eff}} = \kappa$. Dense marks whose piecewise-linear T approaches $t(\tau - t)$ attain the bound in the limit: interior $u_j S_j$ share one sign and are balanced by $\delta_j |u_j| = \Delta_j |S_j|/m$, while endpoint marks become sharp with recoils invisible to the invariant statistic. Finite dense approximants preserve the trade-off. The constant is a limiting sharp constant, not an attained finite-mark minimum.

Quantum Gaussian probes and linear transport reproduce this statistic's Gaussian law through positive probe Wigner covariances. The body state cancels from every invariant statistic and need not be Gaussian. If $|\rho_j| \le \rho < 1$, the quantum covariance inequality gives

$$\kappa_{\text{eff}} \ge \frac{\hbar}{2}\sqrt{\frac{1-\rho}{1+\rho}}, \qquad \boxed{\tau \Delta E \ge 24 z_{1-\epsilon}^2 \hbar \sqrt{\frac{1-\rho}{1+\rho}}.}$$

Three marks at $0, \tau/2, \tau$ with statistic $R_1 - 2R_2 + R_3$ approach within $4/3$ in action. The signal is $F\tau^2/(4m)$; outer recoils cancel. For the middle probe, balance $2\delta = \tau\Delta/(2m)$ and choose the favourable $\rho > 0$. The variance becomes $(\tau\hbar/m)\sqrt{(1-\rho)/(1+\rho)}$, giving the sufficient

limiting threshold $32z_{1-\epsilon}^2\hbar\sqrt{(1-\rho)/(1+\rho)}$. Outer marks are arbitrarily sharp in this limit. The threshold tends to zero as $\rho \uparrow 1$; correlation control, not covariance area alone, is required.

These are Gaussian, invariant, preparation-ignorant tests. A known contractive preparation focused at the final time can escape using one sharp readout; non-Gaussian real grid probes can escape even with $\rho = 0$, paying diverging recoil. Their exact construction and the preparation-aperture counterexample are in general bounds. Yuen's contractive-state mechanism (metadata) is an established comparison, not a new literature claim.

Forgotten Gaussian records and the refinement limit

For Gaussian position Kraus operators

$$M_y(x) = (2\pi\sigma^2)^{-1/4} e^{-(x-y)^2/(4\sigma^2)},$$

forgetting the outcome multiplies the density matrix by $e^{-(x-x')^2/(8\sigma^2)}$, because direct integration of $M_y(x)M_y(x')$ completes the square. Fourier transformation to the Wigner function gives momentum convolution with a centred Gaussian of variance $v = \hbar^2/(4\sigma^2)$. Position marginals are unchanged. Concurrent precisions and variances add, while free/constant-force transport shears a kick at t_j to final displacement $((T-t_j)\delta p/m, \delta p)$. The final noise covariance is therefore

$$\sum_j v_j \begin{pmatrix} (T-t_j)^2/m^2 & (T-t_j)/m \\ (T-t_j)/m & 1 \end{pmatrix}.$$

For a nested family of retained kicks, $\sum_j \sigma_j^{-2} < \infty$ makes this matrix converge by absolute domination. A divergent sum gives divergent momentum variance and no normal momentum-probability limit. For general triangular refinement arrays, convergence of total precision alone is insufficient: all three displayed weighted moments must converge. At the regular rate $\sigma_j^{-2} = \kappa\tau_j$, Riemann sums give the finite continuous-measurement covariance and momentum diffusion rate $\hbar^2\kappa/4$, with density-matrix generator $-\kappa[x,[x,\rho]]/8$. Free-motion Fisher information about initial place with known momentum is $\sum_j \sigma_j^{-2}$. Harmonic forces rotate kicks and require the corresponding transport matrix instead.

This channel and continuous limit are standard (Caves–Milburn 1987, metadata). The comparison with a gauge parallel move is structural: an observed insertion adds a pointer and forgetting leaves a channel, whereas an unobserved time insertion is a different operation. In the classical zero-recoil analogue, Bayesian updating followed by forgetting returns the prior. That identity does not establish quantum kinematics or select $\hbar$.

Consequence for STATE

The classical interval theorem, Gaussian invariant theorem and supplied quantum probe uncertainty are maintained with their distinct assumptions. Post-record restrictions, preparation access, Gaussianity and correlation bounds cannot be silently exchanged. The positive action parameter in the quantum statements is supplied physical $\hbar$; classical completion, quantum instruments, universality and scale identification remain separate.